MONER-JUI • TECHNICAL ARCHITECTURE & STACK

Gentle Dementia & Memory Care Web Application | Project Technical Specification

Category	Technology / Libraries	Architecture & Purpose
Frontend & UI / UX	**HTML5, Modern CSS3, JavaScript (ES6+), Canvas Confetti**	Elder-first responsive design, warm rose/pink calming palette, high-contrast accessible typography, zero countdown pressure.
Backend Architecture	**Serverless Functions, Vercel Edge Network, RESTful APIs**	Scalable serverless execution, fast edge delivery, automatic HTTPS, zero-maintenance global CDN hosting.
Database & Realtime Engine	**Supabase (PostgreSQL), Realtime WebSockets, LocalStorage**	Cloud relational database for users, game sessions, and 8 daily tasks with instantaneous live WebSocket updates and offline fallback.
Speech & Audio Engine	**Web Speech API (SpeechSynthesis), Web Audio API (Synthesizer)**	Gentle female voice assistant (on-demand guidance in English) paired with harmonic solfeggio chimes (528Hz) for cognitive stimulation.
Data Analytics & Visualization	**Chart.js Data Engine**	Interactive statistical bar graphs (performance across Levels 1-3) and doughnut charts (daily task completion rates) on Host Dashboard.
Authentication & Security	**Row Level Security (RLS), Role-Based Access Control (RBAC)**	Secured patient login vs administrative host portal with patient profile deletion and log clearing privileges.
DevOps & CI / CD	**Git, GitHub, Vercel Continuous Deployment**	Automated build, test, and zero-downtime deployment pipeline linked directly to GitHub repository.

Key Technical Highlights

• **3 Difficulty Levels in Cognitive Games:** Memory Match (Visual Pattern Recall) and Remember the Order (Sequential Recall) calibrated across Level 1 (Easy), Level 2 (Medium), and Level 3 (Mastery).

• **8 Daily Wellness Tasks Tracker:** Live-synced checkboxes monitoring hydration, medication, light exercise, and family connection with real-time feedback to the host.

• **Real-Time Caregiver Dashboard:** Central host portal tracking multi-patient activity, live game logs, time spent, accuracy percentages, and aggregated scores.

• **Safe Storage Management:** Host controls to permanently delete inactive patient profiles or flush historical session logs directly from the cloud database.

Live Prototype URL: https://monerjuiprototype.vercel.app

GitHub Repository: https://github.com/jaysteam04072003-cell/Moner-Jui

Host Username: Abhishek Kawarkhe

Host Password: xxxxx